Department of Computer Science

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Project Title: **Undo/Redo Program**

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1. Problem Statement

When users interact with text editors, they frequently make unintentional edits, delete important content, or change text accidentally. Without an undo/redo feature, these mistakes become difficult or impossible to reverse.
This project aims to solve that issue by designing a simple editing system that tracks user actions and allows them to revert or reapply those actions.
The purpose is to build a small-scale model that demonstrates how stacks support action history in a real-world application.

2. Objectives

- To implement a functional undo/redo mechanism using stacks.
- To store each user action and restore previous states when needed.
- To apply the **Last-In-First-Out (LIFO)** concept practically.
- To simulate core text-editor behavior in a beginner-friendly form.
- To strengthen understanding of stack operations such as push and pop.
- To produce a clean interface for testing different operations.

3. Features / System Scope

- Add new text to the editor
- Remove previously entered text
- Undo the most recent action using stack operations
- Redo the last undone action
- Maintain two stacks:
    - One for undo
    - One for redo
- Clear redo history after any new action
- Show the current state of text after every update
- Handle cases where undo/redo is not possible
- Provide a simple menu-driven interface

4. Tools & Technologies

- **Language:** C++
- **Environment:** VS Code / Dev C++
- **Data Structure Used:**
    - **Stacks** (primary structure)
    - Strings/dynamic arrays (to store text)

5. Expected Results

The final program will enable the user to:

- Enter and delete text
- Reverse actions through Undo
- Reapply reversed actions using Redo
- Observe how stacks can maintain a history of operations
- See updated text after each modification
- Understand clearly how LIFO logic is used to track user activity

The project will demonstrate a practical application of the stack data structure in software development.

6. Project Timeline (Optional)

Stage	Task
Week 1	Select topic & prepare proposal
Week 2	Design system logic & action structure
Week 3	Implement basic text handling
Week 4	Add Undo functionality
Week 5	Implement Redo functionality
Week 6	Debug, refine, and test
Week 7	Prepare report + diagrams
Week 8	Final demo & submission